

Predictive Analysis:

1. Predictive analysis is a type of data analysis that uses statistical methods and machine learning techniques to predict future outcomes based on past data. It helps organizations make better decisions, forecast future trends, and reduce risks.

2. The main goal of predictive analysis is to find patterns and relationships in historical data. These patterns are then used to make accurate predictions about what might happen in the future.

3. Working: It involves;

i. Collecting data from different sources.

ii. Cleaning and preparing the data.

iii. Applying algorithms and mathematical models.

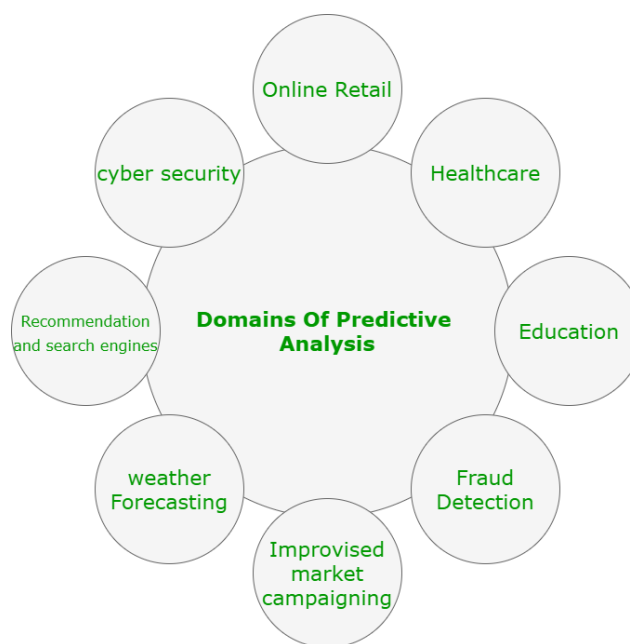
iv. Identifying trends and making predictions.

4. Challenges:

i. Need for Large Data: Predictive analysis requires a lot of historical data. This can be difficult for organizations that are new to using this technique.

ii. Data Quality: The data must be accurate and relevant. Poor-quality data can lead to incorrect predictions.

iii. Choosing the Right Model: Each model has its own pros and cons. The best model depends on the type of data and the problem you are trying to solve. There are many models to choose from, like: Linear Regression, Decision Trees, Neural Networks.

**Stages of Predictive analysis:**

1. Descriptive Analysis: Focuses on understanding *what happened in the past*. It summarizes historical data to identify patterns, trends, or KPIs. **Example: Sales increased by 20% last quarter compared to the previous one.**

2. Diagnostic Analysis: This type of analysis digs deeper to understand why certain patterns or events occurred. It often uses techniques like data drilling, correlation analysis, and root cause analysis. **Example: Finding out why customer churn increased in a specific month.**

3.Predictive Analysis: Predictive analysis uses statistical models and machine learning to forecast future outcomes based on past data. It is a key part of data mining that helps make informed decisions. **Example: Predicting which customers are likely to stop using a service.**

4.Prescriptive Analysis: This stage goes beyond prediction to recommend specific actions to achieve desired outcomes. It often uses optimization and simulation techniques. **Example: Suggesting marketing strategies for retaining high-value customers.**

5.Cognitive Analysis: Cognitive analysis uses AI and machine learning to simulate human thought processes. It enables systems to learn from data, interpret natural language, and make autonomous decisions. **Example: A virtual assistant that analysis user behaviour and recommends personalized products.**

Population And Sample: In data mining, it's often too hard or expensive to study all the data (the whole population). So, we use a smaller part called a sample. If the sample is chosen well, it can still give us accurate results. This helps us build models, test ideas, and find patterns without needing all the data.

1.Population: The population refers to the entire set of data or all possible observations relevant to a particular study or analysis. It includes every individual or item that fits a certain criteria. **Example: All customers of an e-commerce website over the past 5 years.**

2.Sample: A sample is a subset of the population that is selected for analysis. It is used when analyzing the whole population is too costly or time-consuming. The goal is to make conclusions about the whole population by studying the sample. **Example: A group of 1,000 randomly selected customers from the e-commerce website.**

Measure of Central Tendency: Measures of central tendency are statistical techniques used in data mining to identify the center or average value of a dataset. They provide a summary of large data by showing the typical or most representative value.

1.Mean: The most common measure is the mean, calculated by adding all data values and dividing by the total number of values. It is best used for numerical data without extreme outliers.

2.Median: The median is the middle value in a sorted list of numbers. It is especially useful when dealing with skewed data or when outliers may distort the mean.

3.Mode: The mode represents the value that appears most frequently in a dataset. It is suitable for both numerical and categorical data, especially when identifying the most common category or value.

4.Importance: These measures help in data understanding, cleaning, and preprocessing. They are crucial for tasks such as filling missing values, comparing datasets, and detecting patterns or anomalies during analysis.

5.Ex: In a company, most employees earn between ₹25,000 to ₹35,000 per month, but a few top managers earn over ₹1,00,000. When analyzing the salary data:

- i. **The mean salary may be quite high due to the few very high salaries, even though most employees earn less.**
- ii. **The median salary, being the middle value, gives a better idea of what a typical employee earns.**
- iii. **The mode salary may show the most commonly earned amount among staff, like ₹30,000.**

Measure of Dispersion: Measures of dispersion describe the spread or variability in a dataset. In data mining, they help understand how much the data values deviate from the central tendency (like the mean or median).

1.Range: The difference between the largest and smallest values in a dataset.

Formula: Range = Maximum value – Minimum value

2.Variance: The average of the squared differences from the mean. It shows how data points differ from the average value.

Formula (for sample variance): Variance (s^2) = $(x_1 - \text{mean})^2 + (x_2 - \text{mean})^2 + \dots + (x_n - \text{mean})^2 / n - 1$

3.Standard Deviation: The square root of the variance. It shows how much values deviate from the mean, in the same unit as the data.

Formula: Standard Deviation (s) = $\sqrt{\text{Variance}}$

4.Ex: a = 10, 12, 14

Step 1: Calculate the Range:

Range = 14 - 10 = 4

Step 2: Calculate the Mean ($\bar{x}$):

$\bar{x} = (10 + 12 + 14) / 3 = 36 / 3 = 12$

Step 3: Calculate the Variance (s^2):

$s^2 = [(10 - 12)^2 + (12 - 12)^2 + (14 - 12)^2] / (3 - 1)$

$s^2 = [(-2)^2 + (0)^2 + (2)^2] / 2$

$s^2 = [4 + 0 + 4] / 2 = 8 / 2 = 4$

Step 4: Calculate the Standard Deviation (s):

$s = \sqrt{4} = 2$

What are measures of Shapes (Skewness and Kurtosis):

Skewness: Skewness is a key statistical measure that shows how data is spread out in a dataset. It tells us if the data points are skewed to the left (negative skew) or to the right (positive skew) in relation to the mean. It is important because it helps us to understand the shape of the data distribution which is important for accurate data analysis and helps in identifying outliers and finding the best statistical methods to use for analysis.

Types of Skewness:

1.Positive Skewness (Right Skew): In a positively skewed distribution, the right tail is longer than the left which means most data points are on the left with a few large values pulling the distribution to the right.

Relationship:

$\text{Mean} > \text{Median} > \text{Mode}$

Examples: Income distribution, exam scores and stock market returns.

2.Negative Skewness (Left Skew): In a negatively skewed distribution, the left tail is longer which means most data points are on the right with a few smaller values pulling the distribution to the left.

Relationship:

$\text{Mean} < \text{Median} < \text{Mode}$

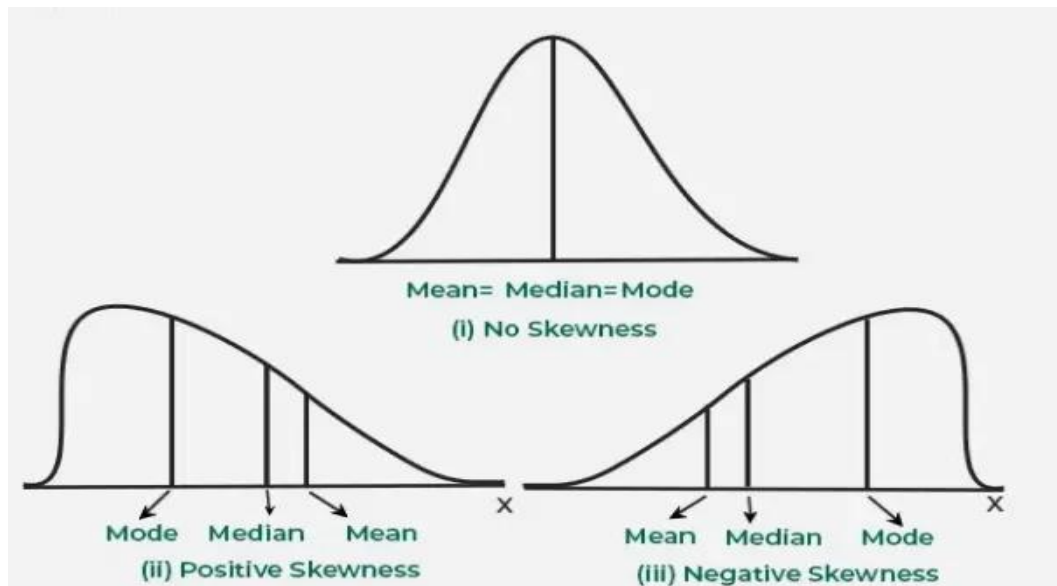
Examples: Test scores on easy exams, age at retirement and gestational age at birth.

3.Zero Skewness (Symmetrical Distribution): Zero skewness shows a perfectly symmetrical distribution where the mean, median and mode are equal. In a symmetrical distribution, the data points are evenly distributed around the central point.

Relationship:

$\text{Mean} = \text{Median} = \text{Mode}$

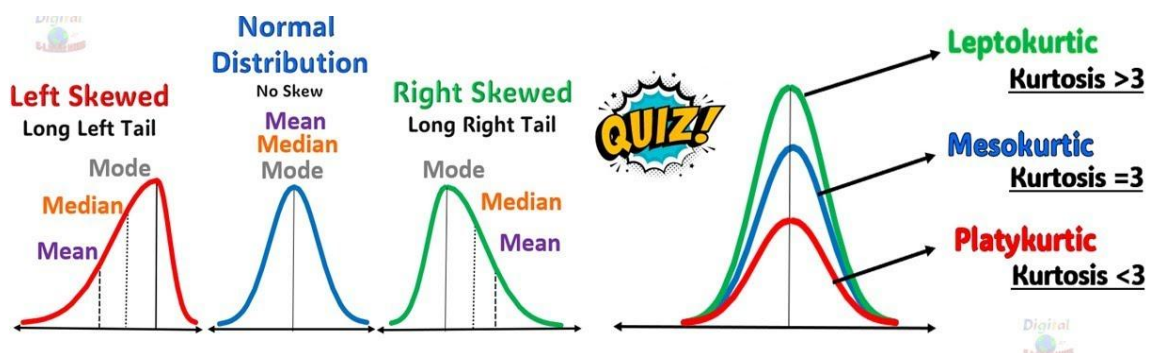
Example: A perfectly balanced dataset with equal frequencies of all values.



Kurtosis: Kurtosis in data mining is a statistical measure that describes the shape of a data distribution, focusing on how the tails and the peak compare to a normal distribution. It is mainly used to understand whether a dataset has more or fewer extreme values than expected in a normal curve.

Types:

- 1. Leptokurtic:** A leptokurtic distribution has a kurtosis value greater than 3, meaning the curve is more peaked with heavier tails, showing that the data contains more extreme values or outliers.
- 2. Mesokurtic:** A mesokurtic distribution has a kurtosis value close to 3, which indicates a normal bell-shaped curve with moderate tails and no extreme outliers.
- 3. Platykurtic:** A platykurtic distribution has a kurtosis value less than 3, resulting in a flatter curve with lighter tails and fewer extreme values.



Sampling Distribution: A sampling distribution refers to a probability distribution of a statistic that comes from choosing random samples of a given population. Also known as a finite-sample distribution, it represents the distribution of frequencies on how spread apart various outcomes will be for a specific population. The sampling distribution depends on multiple factors – the statistic, sample size, sampling process, and the overall population. It is used to help calculate statistics such as means, ranges, variances, and standard deviations for the given samples.

Working:

1. Select a random sample of a specific size from a given population.
2. Calculate a statistic for the sample, such as the mean, median, or standard deviation.
3. Develop a frequency distribution of each sample statistic that you calculated from the step above.
4. Plot the frequency distribution of each sample statistic that you developed from the step above. The resulting graph will be the sampling distribution.

Central Limit Theorem: The Central Limit theorem says that if we take many random samples from any population and calculate their averages, those averages will form a bell-shaped (normal) curve even if the original data is not normally distributed as long as the sample size is large enough. This helps us make predictions about the whole population using just sample data. By calculating sample means these averages will tend to form a normal distribution. This normality holds true as long as the sample size is sufficiently large, typically $n \geq 30$ providing the foundation for making inferences about populations even when we don't have access to all the data.

Steps:

1. Pick random samples from the population.
2. Find the average of each sample.
3. Make a graph of all the sample averages.
4. Notice the shape, which becomes normal if samples are big enough.
5. Remember: the average of these means is close to the real population average.

Confidence Interval: A confidence interval gives us a range of values that we believe to be true average (mean) of something is likely to be in.

Ex: Suppose we want to estimate the average marks of all students in a school. We take a random sample of 50 students and find that their average marks are 72. Using the sample data, we calculate a 95% confidence interval and find that the true average lies between 68 and 76. This means we are 95% confident that the actual average marks of all students in the school fall between 68 and 76.

Hypothesis Testing: Hypothesis testing compares two opposite ideas about a group of people or things and uses data from a small part of that group (a sample) to decide which idea is more likely true. We collect and study the sample data to check if the claim is correct.

Ex: If a company says its website gets 50 visitors each day on average, we use hypothesis testing to look at past visitor data and see if this claim is true or if the actual number is different.

1. Null Hypothesis (H_0): The starting assumption. For example, "The average visits are 50."

2. Alternative Hypothesis (H_1): The opposite, saying there is a difference. For example, "The average visits are not 50."

Working of Hypothesis Testing:

Step 1: Define Hypotheses:

- **Null hypothesis (H_0):** Assumes no effect or difference.
- **Alternative hypothesis (H_1):** Assumes there is an effect or difference.

Ex: Test if a new algorithm improves user engagement.

Note: In this we assume that our data is normally distributed.

Step 2: Choose significance level: We select a significance level (usually 0.05). This is the maximum chance we accept of wrongly rejecting the null hypothesis (Type I error). It also sets the confidence needed to accept results.

Step 3: Collect and Analyze data:

- Now we gather data this could come from user observations or an experiment. Once collected we analyze the data using appropriate statistical methods to calculate the test statistic.
- **Ex:** We collect data on user engagement before and after implementing the algorithm. We can also find the mean engagement scores for each group.

Step 4: Calculate Test Statistic:

The test statistic measures how much the sample data deviates from what we did expect if the null hypothesis were true. Different tests use different statistics:

- **Z-test:** Used when population variance is known and sample size is large.
- **T-test:** Used when sample size is small or population variance unknown.
- **Chi-square test:** Used for categorical data to compare observed vs. expected counts.

Step 5: Make a Decision: We compare the test statistic to a critical value from a statistical table or use the [p-value](#):

1.Using Critical Value:

- If test statistic > critical value → reject H_0 .
- If test statistic $\leq$ critical value → fail to reject H_0 .

2.Using P-value:

- If p-value $\leq \alpha$ → reject H_0 .
- If p-value > α → fail to reject H_0 .

Ex: If p-value is 0.03 and α is 0.05, we reject the null hypothesis because $0.03 < 0.05$.

Step 6: Interpret the Results: Based on the decision, we conclude whether there is enough evidence to support the alternative hypothesis or if we should keep the null hypothesis.

Real life Examples of Hypothesis Testing: A pharmaceutical company tests a new drug to see if it lowers blood pressure in patients.

Data:

- **Before Treatment:** 120, 122, 118, 130, 125, 128, 115, 121, 123, 119
- **After Treatment:** 115, 120, 112, 128, 122, 125, 110, 117, 119, 114

One tail test: A one-tailed test is a statistical test in which the critical area of a distribution is one-sided so that it is either greater than or less than a certain value, but not both. If the sample being tested falls into the one-sided critical area, the alternative hypothesis will be accepted instead of the null hypothesis.

Two tail test: A two-tailed test, in statistics, is a method in which the critical area of a distribution is two-sided and tests whether a sample is greater than or less than a certain range of values. It is used in null-hypothesis testing and testing for statistical significance. If the sample being tested falls into either of the critical areas, the alternative hypothesis is accepted instead of the null hypothesis.

Accepting or rejecting the null hypothesis in hypothesis testing:

Using p-value: This is the simplest and most common approach.

Rule (applies to all tests):

- If $p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$ (enough evidence against H_0)
- If $p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$ (not enough evidence)

Example:

$\alpha = 0.05$, $p\text{-value} = 0.03 \rightarrow \text{Reject } H_0$

$\alpha = 0.05$, $p\text{-value} = 0.08 \rightarrow \text{Fail to reject } H_0$

Categorical data: The data that represents **qualities or characteristics** rather than numeric measurements. It consists of values that **belong to distinct groups or categories**, and you can usually **name or label** them, but you **cannot meaningfully add, subtract, or average** them.

Chi-Square Test: The Chi-square (χ^2) test is a statistical test used to check whether there is a significant relationship between categorical variables or whether observed data matches expected data. It's widely used in data mining, machine learning preprocessing, and hypothesis testing when working with categorical data.

Types of Chi-square Tests:

1. Chi-square test of independence: Checks if **two categorical variables** are related. Ex: Is gender related to product preference?

2. Chi-square goodness-of-fit test: Checks if **observed frequencies** match an **expected distribution**. Ex: Do dice rolls follow a uniform distribution?

Formula: $X^2 = \sum (O - E)^2 / E$

Where:

- O = Observed frequency
- E = Expected frequency (based on null hypothesis)

Decision:

If $X^2(\text{calculated}) > X^2(\text{critical}) \rightarrow \text{Reject } H_0$

Or

If $p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$

Use in Data Mining:

1. Feature selection: Find important categorical features in classification.

2. Market basket analysis: Identify association between items.

3. Survey analysis: Detect relationship between demographics and preferences.

ANOVA Test: The ANOVA test (Analysis of Variance) is a statistical method used to determine whether there are significant differences between the means of three or more groups. In data mining and machine learning, ANOVA is often used for feature selection and for checking if a categorical variable has a significant effect on a numerical variable.

Working:

ANOVA compares variance between groups to variance within groups using the F-test.

Formula for F-statistic: $F = \text{Variance between groups (MSB)} / \text{Variance within groups (MSW)}$

Where: MSB = Mean Square Between groups, MSW = Mean Square Within groups

Decision Rule:

1. Choose significance level ($\alpha = 0.05$).
2. Calculate **F-statistic**.
3. Find **F-critical** from F-distribution table (based on degrees of freedom).
4. Decision:
If $F_{\text{calculated}} > F_{\text{critical}} \rightarrow \text{Reject } H_0$ (means are not all equal).
Or, if $p \leq \alpha \rightarrow \text{Reject } H_0$.